MIL-L-51196(MU)
14 October 1964
(See section 6)

MILITARY SPECIFICATION

LIGHT, ILLUMINATING AND ULTRAVIOLET

1. SCOPE

1.1 This specification covers one type of ultraviolet (UV) light source with an attached viewing box.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

DL5-77-740 - Light, Illuminating and Ultraviolet.

(All drawings referenced in this specification are listed on Drawing DL5-77-740.)

FSC 6665

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-77-740 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the ultraviolet (UV) light shall conform to the specification and drawings listed on Dwg DL5-77-740 and subsidiary drawings.

3.2 Manufacture and assembly. The UV light shall be manufactured and assembled as specified on Drawing D5-77-740.

3.3 Watertightness. The UV light shall be watertight. No water shall enter the light when tested as specified in 4.6.1.

3.4 Light projection. The UV light shall project a concentrated beam of light such that the projected spot does not exceed 9 inches in diameter on a screen 5 feet away, when tested as specified in 4.6.2.

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3.5 Operation. The UV light shall function properly and the light shall not go off when shaken vigorously when tested as specified in 4.6.3.

3.6 Preproduction sample. When specified in the contract or purchase order, a preproduction sample of the UV lights shall be produced for examination and test. The preproduction sample shall be manufactured using the same materials, methods and equipment as will be used during regular production. (See 6.2.)

3.7 Workmanship. The UV light shall be free of damage (breakage, scratches, holes, or mars), defects (flash, checks, cracks, or blisters) and foreign matter (grease, oil, or viscous material) which could affect the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedures which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2.3 Inspection substation. If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL-STD-105 are applicable to each substation and to the end item.

4.3 Lotting. A lot shall consist of the UV lights produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of molding plastic, UV filters, and lamps shall be represented in any one lot of finished UV lights.

4.4 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For Major A defects, level S-3 shall be used.

4.5 Inspection provisions.

4.5.1 Examination and nondestructive tests. Sample UV lights shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.5.2 Classification of defects.

4.5.2.1 Light, illuminating and ultraviolet (Drawing D5-77-740).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Operation failure	4.6.3
102	Light projection not as specified	4.6.2
103	Component missing or incorrectly assembled	
104	Workmanship (3.7)	
105	Threads incorrect	
106	Lens mounted incorrectly	
Major A:	AQL 1.0 percent defective	
150	Watertightness	4.6.1

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4.6 Tests.

4.6.1 Watertightness. The UV light minus the viewing box shall be charged with batteries and submerged for a period of one hour to a depth of 3 feet ± 1 inch below the surface of a 10 percent solution of sodium chloride at room temperature. The outside of the light shall be wiped dry, disassembled and examined for evidence of water penetration.

4.6.2 Light projection. The UV lights shall be tested using a screen of suitable size which is perpendicular to the optical axis of the light.

4.6.3 Operation. Three Mallory type TR-136 R batteries shall be inserted in the lights. The push button on the switch shall be depressed five times and then the light shall be shaken vigorously with the push button held down to determine compliance with 3.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The UV lights shall be packaged to afford adequate protection from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. UV lights, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The UV light covered by this specification is intended for use as an ultraviolet light source which activates fluorescent markings on items in the M19 Sampling and Analyzing Kit. The light is also provided with a viewing box for viewing detector tube reactions at night.

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6.2 Ordering data. Procurement documents should specify the following:

(a) The title, number, and date of this specification.
(b) Whether a preproduction sample is required.

(1) Quantity required.
(2) Time required for supplier submission of samples for Government test and evaluation after award of contract.
(3) Name and address of test facility and shipping instructions when testing is performed by the Government.
(4) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Supersession data. This specification supersedes Chemical Corps Purchase Description 197-54-725, dated 21 May 1959.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 6665-A120

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